



An investigation into the effects of pore connectivity on T_2 NMR relaxation



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ABSTRACT

Nuclear Magnetic Resonance (NMR) is a powerful technique used to characterize fluids and flow in porous media. The NMR relaxation curves are closely related to pore geometry, and the inversion of the NMR relaxometry data is known to give useful information with regards to pore size distribution (PSD) through the relative amplitudes of the fluids stored in the small and large pores. While this information is crucial, the main challenge for the successful use of the NMR measurements is the proper interpretation of the measured signals. Natural porous media patterns consist of complex pore structures with many interconnected or “coupled” regions, as well as isolated pores. This connectivity along the throats changes the relaxation distribution and in order to properly interpret this data, a thorough understanding of the effects of pore connectivity on the NMR relaxation distribution is warranted. In this paper we address two main points. The first pertains to the fact that there is a discrepancy between the relaxation distribution obtained from experiments, and the ones obtained from solving the mathematical models of diffusion process in the digitized images of the pore space. There are several reasons that may attribute to this such as the lack of a proper incorporation of surface roughness into the model. However, here we are more interested in the effects of pore connectivity and to understand why the typical NMR relaxation distribution obtained from experiments are wider, while the numerical simulations predict that a wider NMR relaxation distribution may indicate poor connectivity. Secondly, by not taking into account the pore coupling effects, from our experience in interpreting the data, we tend to underestimate the pore volume of small pores and overestimate the amplitudes in the large pores. The role of pore coupling becomes even more prominent in rocks with small pore sizes such as for example in shales, clay in sandstones, and in the microstructures of carbonates.

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1. Introduction

Nuclear Magnetic Resonance (NMR) relaxation measurements play a unique role amongst other logging techniques in that they provide information about the geometry of the porous media, as well as the fluids within the pores. This information is essential to estimate important flow parameters for a reservoirs such as producibility and permeability. Due to the magnetic moments present in hydrogen, the NMR relaxation signals are extremely sensitive to fluid filled pores containing water and hydrocarbons. Hence, the amplitude of the NMR signal is proportional to the fluid content in the rock. The relaxation rate depends primarily on the pore size

and the petrochemical properties of the rock-fluid interface and on the paramagnetic ions present in the fluid phases.

According to the pioneering work of Brownstein and Tarr [2] for uncoupled pore systems, the multi-exponential decay nature of the proton magnetic relaxation is strongly connected to the geometry of the pores. From this work, many authors have invoked the simple convention of assuming a model of isolated pores where each diffusing proton samples only a single pore, in order to interpret the NMR measurement data. The main problem with this assumption is that only totally disconnected pores reflect correctly the amount of fluid in them via the NMR relaxation distributions. However, real porous media consist of multiple connected pores. As a result, this conventional approach for interpreting the NMR data gives misleading information about the quantity of fluid in the pores as the connectivity along the throats changes the spectra. Typically, wider relaxation distributions indicate that pores are not connected well.

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In this work we interpret the NMR T_2 relaxation curves for porous media with multiple interconnected pores. Although NMR signals give unique information about the properties of both the formation and the fluid in the pore space, there are still some misconceptions regarding the distinction between the NMR T_2 relaxation curves for an interconnected pore system and that of individual pores. An important question to ask is how much of the information obtained from NMR data for connected pores reflect on the quantity of fluid in the individual pores and its size and shape. Understanding how this pore coupling effect influences the NMR measurement data is fundamental in geophysical applications. For sufficiently weak connectivity, the corresponding NMR T_2 relaxation amplitudes resemble the T_2 relaxation curves for the isolated pores. This is an important point because a key feature in tight formations such as in shale matrix or clay bound sandstone is the low pore connectivity, where a proper interpretation of the NMR signals can give more accurate information about the formation. Moreover, another important potential application is in the NMR estimates of permeability where again conventional thinking bases these estimates on theoretical models that show an increase in permeability is due to an increase in pore size and porosity only. However, we point out that the NMR signals rely on more than just pore body size distribution as they include also the effect of the pore connectors or throats that can help improve permeability estimates.

Theoretical studies have been done to understand the characteristics of a coupled pore systems by studying the eigenvalues and eigenfunctions that characterize the shape of the pore space. In one particular study Kac [8], looked at the normal modes of vibration of the membrane of a drum and posed the question of whether it is possible to deduce the shape of an individual drum from a multiply connected set of drums. McCall et al. [11] studied the behavior of magnetization in a pore space where an entire range of coupling strengths is considered. Their approach relied on developing perturbation theory solutions to the equations of motion in both the weak and strong coupling limits. Ramakrishnan [14] suggested that a consequence of the diffusion coupling between pores, the assumption that directly connects the T_2 distribution with pore size distribution breaks down in many pore systems, thus limiting our ability to make predictions on permeability and movable fluid fractions. Zelinski [20] studied the effects of diffusion coupling on the higher eigenmodes in a connected pore system.

Experimental and numerical studies have also been performed in order to better understand how NMR measurements are affected by the interconnectedness of pores. Toumelin et al. [18] through laboratory studies and numerical models showed that inter-pore diffusion can strongly influence NMR measurements in carbonates, and that there is a tendency for the amount of pore coupling to increase with temperature as pore-fluid diffusivity increases. Annand and Hirasaki [1] investigated geometric controls on pore coupling in microporous grain packs. They concluded that the coupling between micropores and large intergranular pores become more significant with increasing grain diameter. Laboratory investigation to improve the understanding of geochemical controls on pore coupling, and to better interpret the NMR relaxation measurements in the presence of pore coupling have been performed by Grunewald et al. [6]. Fleury [3] studied pore to pore diffusive exchange using the NMR T_2 - store - T_2 2D experimental technique. Mohnke et al. [12] used microscale simulations of NMR relaxation in porous media to demonstrate the effect of pore coupling on the T_2 distribution for 2D numerical models. More recently, [10] studied the impact of micro-fractures and fracture-pore coupling on NMR based permeability estimates, by synthetically generating two pore-scale images of a fractured carbonate rocks and performing some numerical simulations. Fleury et al.

[4] have done NMR cryoporometry experiments for characterizing shales in their natural hydrate state and did a comparison of their results with the NMR T_2 distribution. They found that their results differ significantly as the T_2 relaxation is influence by the pore coupling.

In order to better understand the impact of the pore connectivity on the NMR spectra and the correlations between the complex porous media and the NMR parameters we rely on both theoretical calculations and on numerical simulations. We present some analytical models of a simple two pore system for different parameters and present theoretical findings in the presence of pore exchange. We then look at some 3-D models where we change the radius of the connectors to explore the effect of throat size on the NMR T_2 relaxation curves. For these models we also do NMR permeability estimates and compare with absolute permeability from flow calculations. Lastly, we discuss these ideas further, and demonstrate the numerical performance in comparison with experiment for two types of porous media. The first porous media is constructed virtually based on a given particle size distribution of the Mountain Silica sand pack, and the second porous media, is a sandstone core plug which is then digitally constructed through μ -CT scans.

2. Theory

NMR spectroscopy is applied to saturated porous media where the nuclei of certain atoms that are already immersed in a static magnetic field B_0 , are then exposed to a second oscillating magnetic field. For petro-physical applications, the pore space is concentrated with water or oil in which in the presence of the magnetic field produces excitation of the protons. NMR relaxation occurs when there is an interaction of the nuclear spins with the bulk fluid and the pore walls. The mathematical model that represents the evolution of the magnetization during the excitation and the resulting NMR relaxation of the protons in the fluid is described by the Bloch-Torrey equations which model diffusion plus a reaction term. For simplicity of presentation we consider here a scalar diffusion equation with a reactive term along with boundary conditions modeling passive and active relaxation at the pore walls.

3. Two pore systems

We consider a two-pore system where Pore 1 and Pore 2 are separated by an interface located at distance α as depicted in Fig. 1. For this simple geometry, the flow is governed by the molecular diffusion where the concentration of energized nuclei is given by c . Moreover, there is a surface like sink strength density σ_s over the active surface s . This results in a decay in the total nuclear magnetization which is given by:

$$S(t) = \int c(x, t) dV.$$

Representing the actual signal from the full sample volume V . In the analysis below we consider only the contribution of those modes having no y or z -dependence.

The diffusion-reaction equation is then given by:

$$\frac{\partial c}{\partial t} - D \frac{\partial^2 c}{\partial x^2} + \frac{c}{T_{2B}} = 0 \quad (1)$$

where the solution is decomposed into two parts as: $c(x, t) = c_1(x, t) + c_2(x, t)$. Here c_1 , c_2 are concentrations in Pore 1 and Pore 2 respectively. D is the scalar diffusion coefficient where we assume to be constant, and T_{2B} is the bulk relaxation time. The boundary conditions are given by:

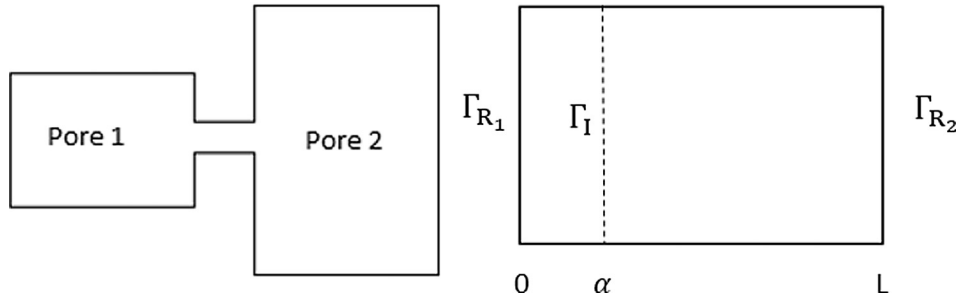


Fig. 1. Two pore system used in the analytical model. Two pores separated by a connector where the interface between the pores is at $x = \alpha$.

$$\begin{cases} D \frac{\partial c_1}{\partial x} - \sigma_s c_1 = 0, & \text{on } \Gamma_{R_1} \\ D \frac{\partial c_1}{\partial x} - D \frac{\partial c_2}{\partial x} = 0, & \text{on } \Gamma_I \\ D \frac{\partial c_1}{\partial x} - \kappa(c_2 - c_1) = 0, & \text{on } \Gamma_I \\ D \frac{\partial c_2}{\partial x} + \sigma_s c_2 = 0 & \text{on } \Gamma_{R_2}, \end{cases} \quad (2)$$

where Γ_{R_1} and Γ_{R_2} are Robin boundary conditions corresponding to the relaxation at the surface, where σ_s is the surface relaxivity and reflects the molecular interactions at the fluid boundaries leading to a loss of magnetization. Here we take $\sigma_s = 5.0 \mu\text{m/s}$. Γ_I represents the interface boundary conditions located at $x = \frac{\alpha}{L}$. The first interface boundary condition enforces continuity of the diffusion flux of first region (Pore 1) across the interface. The second interface describes the jump across the interface through κ which is a non-dimensional number accounting for the size of the connector from Pore 1 to Pore 2. For the initial concentration, we assume that $c(x,0) = S(0)/V$ is constant throughout and is set equal to 1. Now, the complete solution is expressed as:

$$c(\vec{x}, t) = \sum_{n=0}^{\infty} A_n F_n(\vec{x}) e^{-\left(\frac{1}{T_n} + \frac{1}{T_{2B}}\right)t} \quad (3)$$

where $T_n = \frac{L^2}{D\xi_n^2}$, for each eigenvalue ξ_n . $F_n(\vec{x}) = f_1 + (f_2 - f_1)\mathcal{U}_x$, where the unitary operator,

$$\mathcal{U}_x = \begin{cases} 0 & \text{if } x \leq \alpha \\ 1 & \text{if } x > \alpha \end{cases}$$

The coefficients are normalized and are given through orthogonality of the eigenfunctions as:

$$A_n = \frac{\int_0^L F_n(\vec{x}) dx}{\int_0^L F_n^2(\vec{x}) dx}. \quad (4)$$

The amplitude of the total signal will then be given by $\widetilde{A}_n$ and is calculated from the total signal as

$$S(0) = \int_0^L c(x,0) dx = \int_0^L \sum_{n=0}^{\infty} A_n F_n dx = \sum_{n=0}^{\infty} A_n \int_0^L F_n(x) dx = \sum_{n=0}^{\infty} A_n^2 \equiv \widetilde{A}_n.$$

This is a consequence of using the fact that $\int_0^L F_n^2(x) dx = 1$.

Now the general solution for the eigenvalues ξ_n are given by $\mathcal{F}_1(\xi) = 0$, and $\mathcal{F}_2(\xi) = 0$ for Pore 1 and Pore 2 respectively when they are totally independent, corresponding to when $\kappa = 0$, and are given explicitly as:

$$\mathcal{F}_1(\xi) = \xi \tan \frac{\xi \alpha}{L} - \frac{\sigma_s L}{D}, \quad (5)$$

$$\mathcal{F}_2(\xi) = \left(-\xi + \frac{\sigma_s L}{D} \tan \frac{\xi \alpha}{L}\right) \tan \xi + \xi \tan \frac{\xi \alpha}{L} + \frac{\sigma_s L}{D}. \quad (6)$$

The corresponding eigenfunctions are given by

$$\begin{aligned} f_{1n} &= \left(-\xi_n \cos \frac{\xi_n x}{L} + \frac{\sigma_s L}{D \xi_n} \sin \frac{\xi_n x}{L}\right), \quad \text{and} \\ f_{2n} &= \left(\cos \frac{\xi_n x}{L} + \tan \frac{\xi_n \alpha}{L} \sin \frac{\xi_n x}{L}\right). \end{aligned} \quad (7)$$

When $\kappa \neq 0$ the eigenvalues are solved through the transcendental equation, $\mathcal{F}_\kappa(\xi) = 0$ where

$$\mathcal{F}_\kappa(\xi) = \xi \cos \frac{\xi \alpha}{L} \mathcal{F}_1(\xi) \mathcal{F}_2(\xi) - \kappa \left(2\xi \frac{\sigma_s L}{D} + \tan \xi \left[\left(\frac{\sigma_s L}{D}\right)^2 - \xi^2\right]\right), \quad (8)$$

and where the corresponding eigenfunctions for this case are given by

$$f_{1n} = B_n \left(\xi_n \cos \frac{\xi_n x}{L} + \frac{\sigma_s L}{D} \sin \frac{\xi_n x}{L}\right), \quad f_{2n} = \Lambda_n \cos \frac{\xi_n x}{L} + \sin \frac{\xi_n x}{L}, \quad (9)$$

where Λ_n and B_n are defined as

$$\Lambda_n \equiv \frac{\xi_n \cos \xi_n + \frac{\sigma_s L}{D} \sin \xi_n}{\xi_n \sin \xi_n - \frac{\sigma_s L}{D} \cos \xi_n}, \quad \text{and} \quad B_n \equiv \left[\frac{-\Lambda_n \sin \frac{\xi_n \alpha}{L} + \cos \frac{\xi_n \alpha}{L}}{-\xi_n \sin \frac{\xi_n \alpha}{L} + \frac{\sigma_s L}{D} \cos \frac{\xi_n \alpha}{L}} \right]. \quad (10)$$

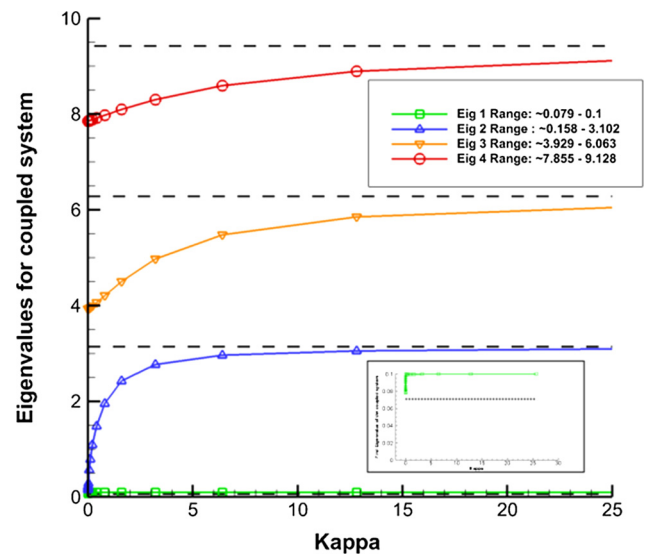


Fig. 2. Eigenvalues for the coupled system. For higher eigenvalues, the dashed lines are multiples of pi. The two lowest eigenvalues are in the green and the blue. The lowest eigenvalues for different Kappa (in green) are magnified and depicted above it. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

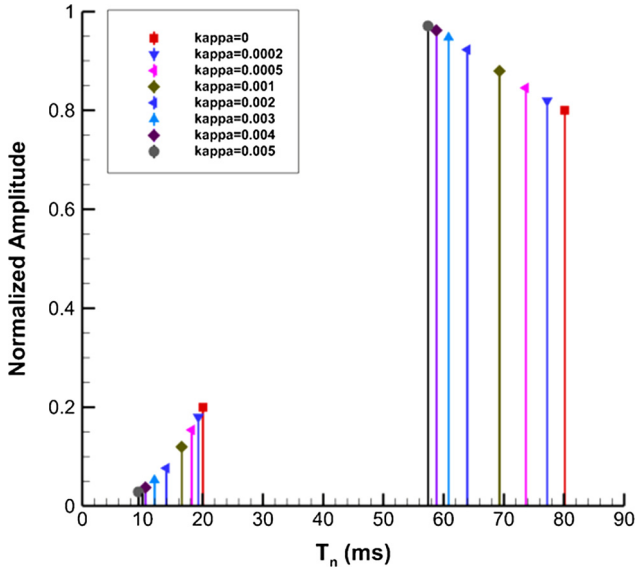


Fig. 3. Normalized amplitude $\tilde{A}_n$ for different T_n by changing κ , which can be considered as the throat diameter. For two disconnected pores ($\kappa = 0$), and weak connections, two distinct peaks can be observed. It is observed that there is a corresponding direction of change with increasing κ . Eventually, when the connection between pores is good then we have a single peak. Here we use $\sigma_s = 10 \mu\text{m/s}$, and $D = 2.07 \times 10^{-9} \text{ m}^2/\text{s}$.

We illustrate the effects of coupling between a small pore and a large pore where we take the interface location $\frac{z}{L} = 0.2$. The relevance of this stems from porous media patterns where there are interconnected pores of varying throat sizes which in this case is designated by κ . The eigenvalues corresponding to $\mathcal{F}_\kappa(\xi) = 0$ for different κ is shown in Fig. 2. NMR relaxation gives multi-exponential decay for multiple pore system and therefore we are only interested in the two smallest eigenvalues. These lowest two eigenvalues are the dominant components in the fast diffusion regime, namely, $\frac{\sigma_s L}{D} \ll 1$ where one eigenvalue describes the change in the larger pore and the other describes the change in the

smallest pore. We are interested in those values of κ that vary from a very weak connection (κ very small) to where the system starts to become well connected. This can be observed from Fig. 2 where for low values of κ there is a substantial change in the eigenvalues.

The coupled system is characterized by the relaxation times T_n that are determined from the dimensionless eigenvalues (where we take for the first two eigenvalues), and decrease as κ increases. This is depicted in Fig. 3 where the relaxation times and the normalized amplitudes are changing with the throat size. For weak connection, for example when $\kappa = 0.0001$, there is not much diffusion between the pores and it closely resembles the $\kappa = 0$ case for isolated pores. However, when the interconnection between pores increases, then the coupled system can be characterized as a single effective relaxation constant with a normalized amplitude of 1.

This suggests that after some time the concentration in the pores begin to relax together. In Fig. 4(a), we consider the case when $\kappa = 0.0005$ and look at the concentration profile of excited protons. It is observed that as they begin to relax, there is a larger jump in concentration of the small pore then the larger pore. However after some time, both pores begin to relax together as the amount of jump in concentration of both pores begins to equalize. From Fig. 4(b) we plot the change in concentration where it is seen that after $t = 0$ the flow begins to become negative which indicates that there is flow from the big pore to the small pore. This flow is being supplied continuously until there is no more change in the concentration (near horizontal line).

Analytically, one can show that by assuming κ is large enough for the system to be considered well connected, then using the fast diffusion approximation, the transcendental equation for the eigenvalues $\mathcal{F}_\kappa(\xi) = 0$ can be approximated to give:

$$\frac{1}{T_n} = \sigma_s \frac{2}{L}, \quad (11)$$

which is the one dimensional version of the surface relaxation $\frac{1}{T_2} = \sigma_s \frac{S}{V}$ where S and V are the surface and volume of the pore. However, it is important to point out here that our relation is for the entire system not for a single pore. That is that once good connection between pores is established and they begin to relax together,

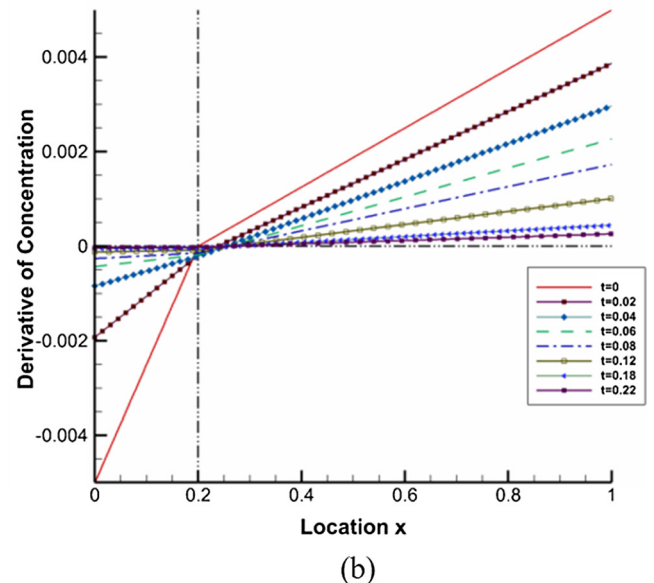
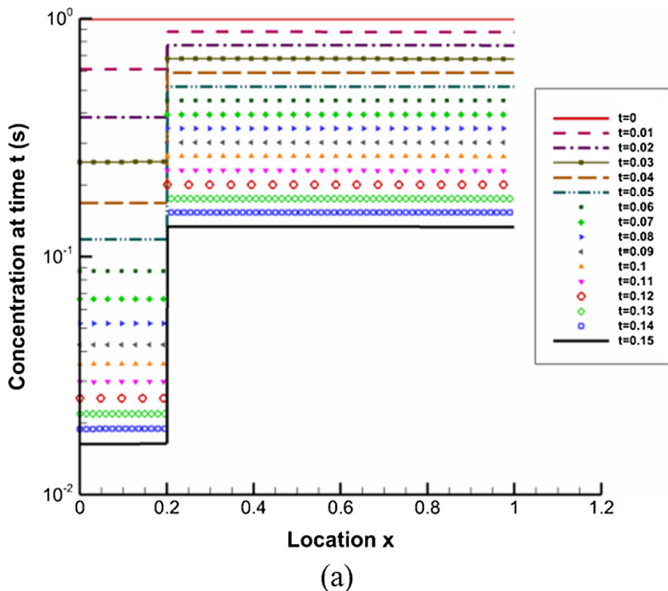


Fig. 4. (a) Depicts the concentration of excited protons for different times where the y axis is in logarithmic scale. This graph shows that when there is weak connection (at earlier times), the jump in the smaller pore is relaxing faster, but when there is better connection between the larger and the smaller pores we see that the two pores start to relax together. (b) Depicts the derivatives of the concentration and how at $t = 0$ it is positive and at $t = 0.02$ s it starts to be negative indicating that the larger pore is supplying flow to the smaller pore.

then the surface to volume ratio obtained must be the average over all the pores.

4. Numerical models

In this section we would like to investigate further using numerical simulations, the effect of pore connectivity on the NMR signals. Therefore, we numerically solve the diffusion-reaction equation with the corresponding boundary conditions for a system representing pores and grains which are depicted as set of concentric spheres. We first construct a system without connectors such that the exact solution will be known and we can verify our numerical results. Fig. 5(a) is an illustration of the 2D cross section of this system where the blue represents fluid in pores and the grey represents solid regions. The central pore has radius of $R = 5 \mu\text{m}$, while all the other spheres in increasing order are calculated to have radii in microns of: $R_1 = 6$, $R_2 \cong 6\frac{1}{6}$, $R_3 = 7$, $R_4 \cong 7\frac{1}{6}$, $R_5 = 8$, $R_6 \cong 8\frac{1}{6}$, $R_7 = 9$, $R_8 \cong 9\frac{1}{6}$, $R_9 = 10$, $R_{10} \cong 10\frac{1}{6}$, $R_{11} = 11$, $R_{12} \cong 11\frac{1}{6}$. The concentric spheres are constructed by imposing two constraints. The first is by ensuring that the T_2 relaxation time for the narrow layers between spheres will be $1/20$ of T_2 for the central sphere of radius R , or

$$\frac{1}{T_2(5 \mu\text{m})} = \frac{1}{20} \left(\frac{1}{T_2(\text{layers})} \right). \quad (12)$$

The explicit relation for the decay rate, T_2 is given by

$$\frac{1}{T_2} = \sigma_s \frac{S}{V} + \frac{1}{T_{2B}},$$

where for small pores or large surface relaxivity, the second term on the right which involves the bulk relaxation time can be neglected and thus for the first layer we have:

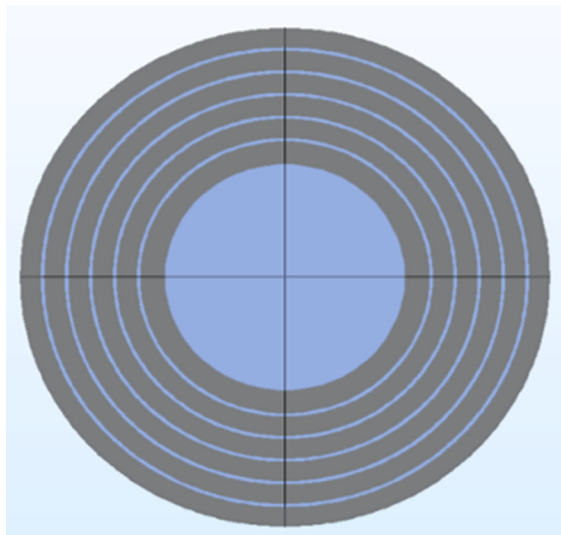
$$\frac{3(R_1^2 + R_2^2)}{R_2^3 - R_1^3} = 20 \frac{3}{R} \Rightarrow \frac{(R_1^2 + R_2^2)}{R_2^3 - R_1^3} = 4. \quad (13)$$

This relation also holds for other layers because the second constraint enforces that the surface to volume ratio of these narrow pore regions are the same. Explicitly written, this leads to the relations:

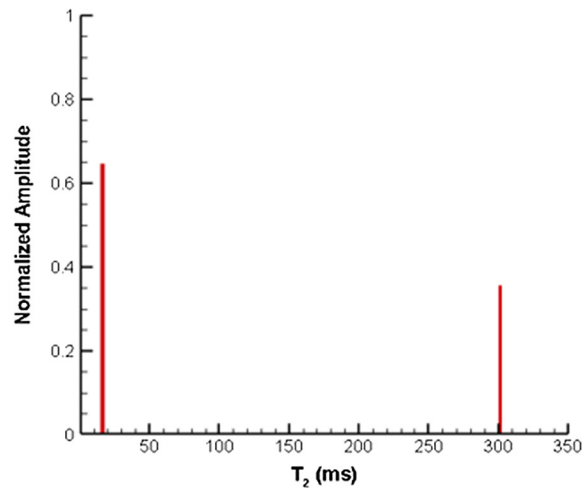
$$\frac{R_2^2 + R_1^2}{R_2^3 - R_1^3} = \frac{R_4^2 + R_3^2}{R_4^3 - R_3^3} = \frac{R_6^2 + R_5^2}{R_6^3 - R_5^3} = \dots \quad (14)$$

By further imposing that $R_1 = R + 1$, $R_3 = R + 2$, $R_5 = R + 3$, $\dots$, we calculate the radii of all the spheres to be that given above. In our numerical simulations we again take $T_{2B} = 3.0 \text{ s}$ and $\sigma_s = 5.0 \mu\text{m/s}$. By construction, this system will lead to the NMR relaxation corresponding to the normalized amplitude vs. T_2 distribution shown in Fig. 5(b). Here T_2 of the larger pores is 0.3 s (as expected) corresponding to a normalized amplitude of 0.3543 and for the smaller pores (layers) is approximately 0.015 s corresponding to a normalized amplitude of 0.6494 . Moreover, as the ratio of the amplitudes is proportional to the ratio of the volumes, we take the sum of the volume of all layers and divide by the volume of the central pores. This gives approximately 1.84 , which verifies our numerical simulation for the case of this system.

Now we discuss the ideas from the previous section by considering connectors that are modeled as cylinders between the different layers and the central pore. There are seven different radii of connectors that range from $0.05 \mu\text{m}$ to $0.6 \mu\text{m}$. We provide connection between the pores as shown in Fig. 6 and we increase the flow between the pores either by increasing the number of connectors or by increasing the radius of the connectors. Under the fast diffusion regime, when there is sufficient flow being supplied from the big pore to the small pores we see from Fig. 7 that T_2 shifts to the left as its amplitude is increasing. This is evident from the fact that as the number of connectors or the connector radii are increased, then more surface is accessible through fast diffusion and the relaxation will become faster. Although, in this case, the amplitude of the smaller pores (or the sum of the volumes of the layers), is initially 1.84 times more than the amplitude in the big pore, and would require substantial amount of flow to be able to shift its T_2 more to the right. This is because the relaxation in the narrow layers happens so quickly that by the time it starts exchanging flow, its concentration has already disappeared. Therefore, as seen in Fig. 7, there is not sufficient amount of flow to cause the left peak to shift to the right. More flow is required to cause the left peak to increase from 16 ms , while the amplitudes are decreasing. In fact when there is sufficient flow, we will obtain one interconnected system where the inversion of the NMR relaxation in



(a)



(b)

Fig. 5. (a) Represents the 2D cross section of the constructed system of pores and grains (concentric spheres) where there is no connectors between the different layers. (b) Depicts the normalized amplitudes vs. T_2 for this system. The numerical inversion of the NMR relaxation is verified by exact solution.

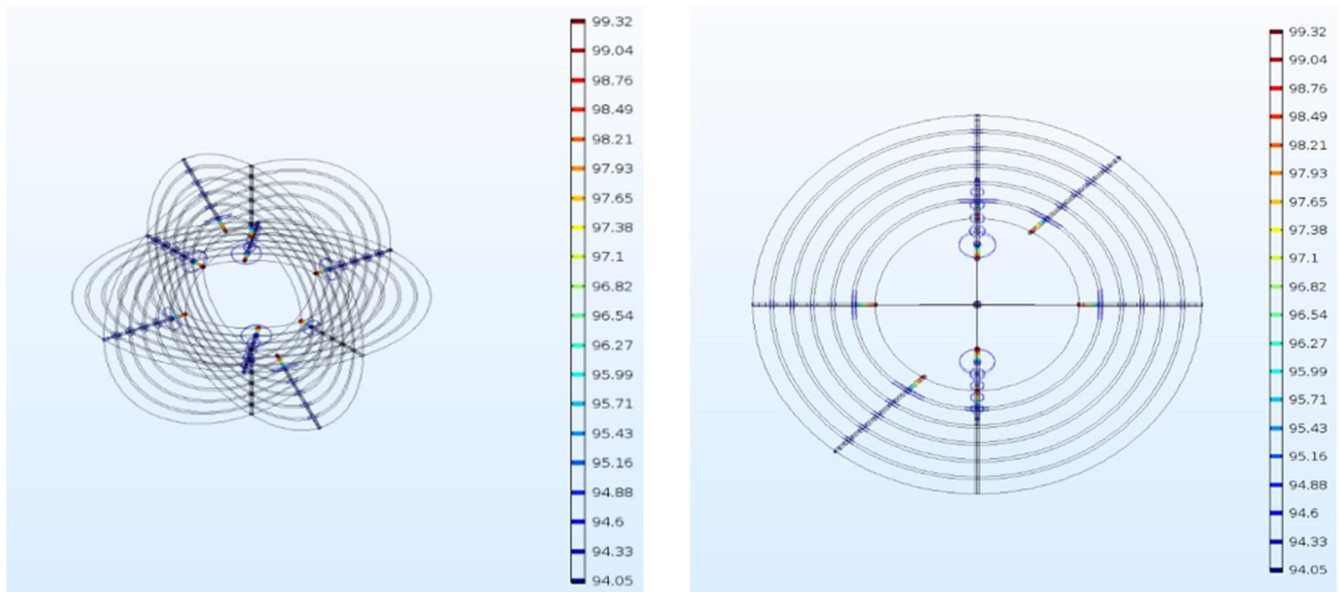


Fig. 6. Contour lines showing the path of diffusion for the connected system. The figure on the right is the 2D view.

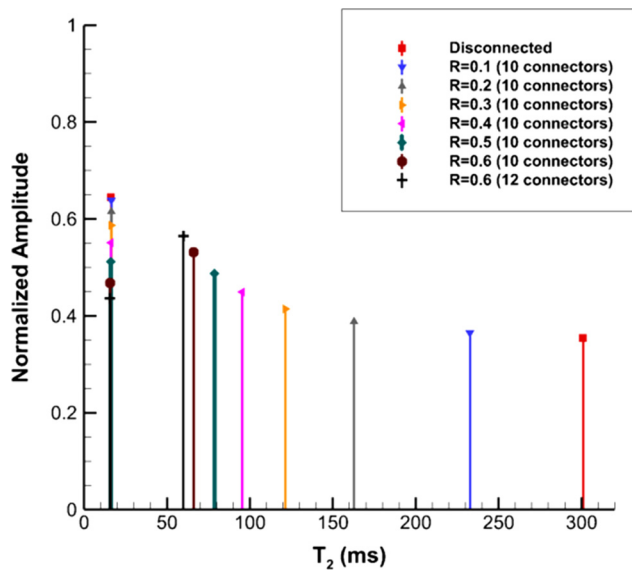


Fig. 7. Here the red peaks correspond to the disconnected system shown in Fig. 5 (b). However, as the connector radii increase we see that the big pore (at 300 ms) is supplying more flow and shifting to the left (T_2 is decreasing) and amplitudes increasing. The small pores (layers) relax very quickly and do not provide supply of flow. The peak amplitudes for the small pores begin to decrease however as the flow from the big pore is the dominating feature the second peaks to not show much shift in T_2 .

this system is represented by a single amplitude. It is easily verified that for this system, the single peak will occur at the T_2 given by

$$\frac{1}{T_2} = \sigma_s \frac{S_1 + S_2}{V_1 + V_2} \quad (15)$$

where S_1 and V_1 are the surface and volume of the big pore respectively, and S_2 and V_2 represent the sum of the surfaces and sum of the volumes of the layers respectively. This is calculated to give $T_2 = 25$ ms.

Furthermore, in order for a connected system to have the same amplitude vs T_2 distribution as a totally disconnected system, the

constructed geometry will be different. If we take the case where the throat radius is $0.6 \mu\text{m}$, as shown in Fig. 8(a) and its NMR amplitude shown in Fig. 8(b), then we can use the fact that the large pore relaxes at $T_{2L} = 67$ ms (0.067 s), to obtain

$$\frac{1}{T_{2L}} = \frac{1}{0.067} = \sigma_s \frac{S}{V} + \frac{1}{T_{2B}} = 5 \frac{3}{R} + \frac{1}{3}$$

which gives $R = 1.028 \mu\text{m}$. In order to construct a disconnected system with the same T_2 of 15.8 ms for the layers obtained from the connector with radius of $0.6 \mu\text{m}$, then we have:

$$\frac{1}{0.0158} = \sigma_s \frac{3(R_2^2 + R_1^2)}{R_2^3 - R_1^3} + \frac{1}{3},$$

also in addition to this, the ratio of the amplitudes in this case is 0.864 leading to the system of equations:

$$\frac{R_2^2 + R_1^2}{R_2^3 - R_1^3} = 4.21 \text{ and } \frac{R_2^3 - R_1^3}{R^3} = 0.864$$

which inevitably leads to the system shown in Fig. 8, where now $R_1 = 1.32 \mu\text{m}$ and $R_2 = 1.48 \mu\text{m}$.

5. Some permeability calculations

In this section we calculate the permeability for the system presented in the last section where we change the connector radii. In order to avoid a simple flow from one end to the other, we force the flow to travel throughout the different layers until reaching outlet, by introducing an extra inclined disc in the center. This will not change the NMR but will make for more realistic flow paths as we changed the throat sizes. In Fig. 9a we show the cross section in the Y-Z plane for this system where the tilted disc is included.

For the flow permeability we solve the Navier-Stokes equations where the flow happens inside the central pore of radius R and the concentric layers. Dirichlet boundary conditions are imposed where we set the pressure at the inlet to be 0.1 Pa, and 0 Pa at the outlet. The connector throats are then varied from $0.05 \mu\text{m}$ to $0.6 \mu\text{m}$. The numerical simulated flow permeability curve is shown in Fig. 9b. Also in Fig. 9b we use the permeability estimate given in [7]

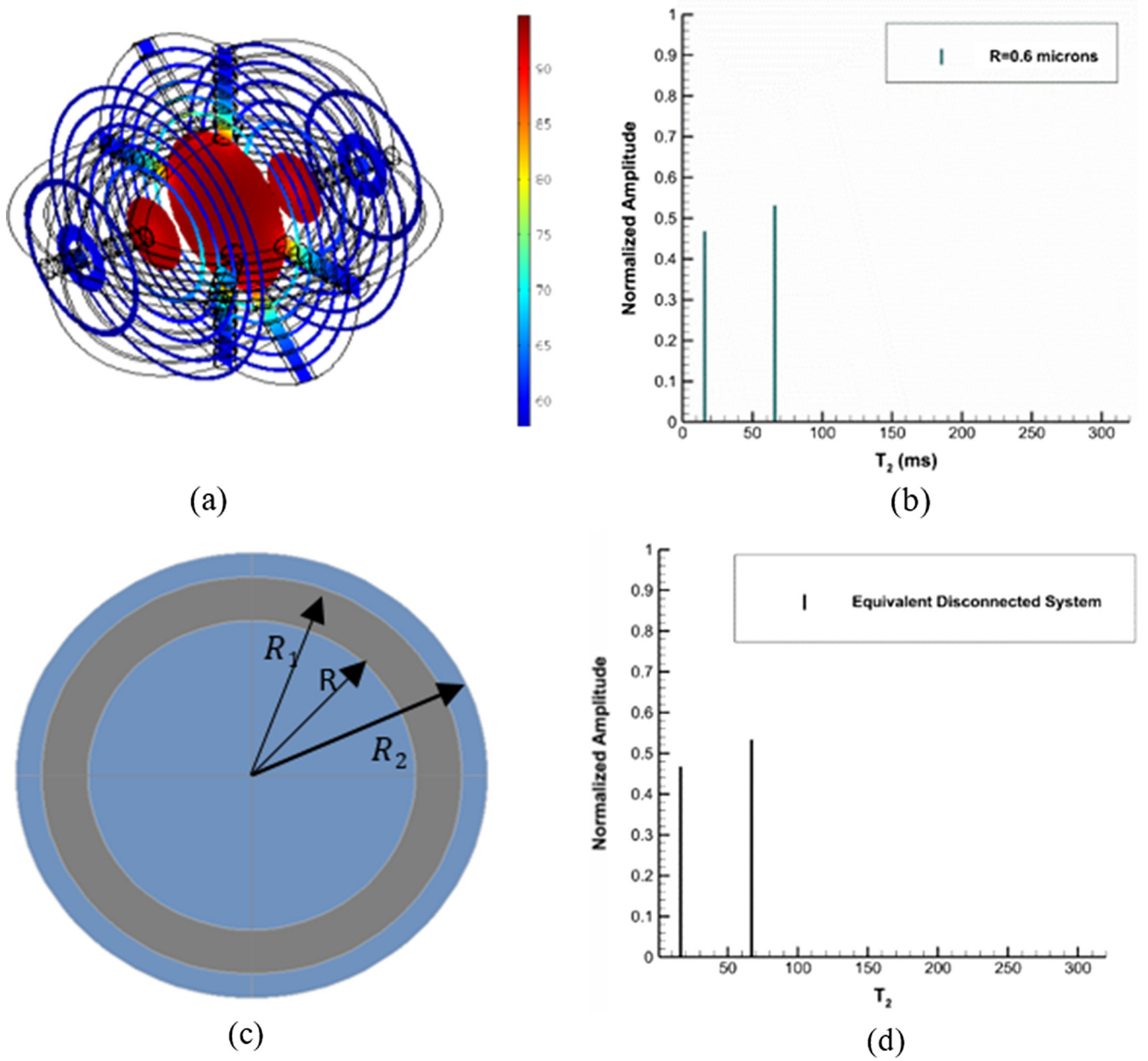


Fig. 8. (a) shows the original 3D geometry used with connectors of 0.6 μm , where the legends depicts the level of concentration of excited protons. (b) Depicts its NMR signature. (c) Demonstrates the disconnected geometry that will give the same NMR signal shown in (d).

$$k_{\text{NMR}} = C \sigma_s^2 \phi^4 (T_{2lm})^2 \quad (16)$$

where C is a constant, and the relaxivity strength σ_s is in $\mu\text{m s}^{-1}$, and the permeability estimate k_{NMR} is measured in mD . In addition, the porosity ϕ for different connector radii does not vary significantly and is approximately 0.26. It is important to also point out that the porosity of the system for different radii of the connectors varies slightly (in the second decimal place). The system is large enough that changing the connector radii does not sufficiently change the porosity. Therefore the porosity is practically constant and thus does not change the concentration (in moles) and NMR amplitude. T_{2lm} represents the logarithmic mean of T_2 distribution as described in [16], and is in milliseconds (ms). The logarithmic mean is defined as

$$T_{2lm} = \exp\left(\frac{\sum_i A_i \ln T_{2i}}{\sum_i A_i}\right). \quad (17)$$

Here C is a fitting parameter that is used to bring the permeability estimate for NMR to scale with the actual permeability results. An arbitrary value 0.0003 was selected so that correlation and computational permeability values fit in the same graph. In this scale it can be readily observed that the permeability estimate is decreasing rather than increasing when the connector radii are increased, which is an incorrect trend. For the permeability calculation Eq. (16), we see that the decreasing T_2 of the big pore causes the permeability estimate to have a decreasing trend. In this model, the surface areas of the layers are very large and the relaxation in the layers occurs very fast. Therefore, from the above analysis, the dominating feature is the relaxation in the big pore and its continuous supply of excited protons. This causes T_2 of the big pore to be decreasing as the connector radii are increasing. As a result, this estimate predicts a decrease in permeability. However, we know that this is not the case, and this estimate does not give the correct predictions.

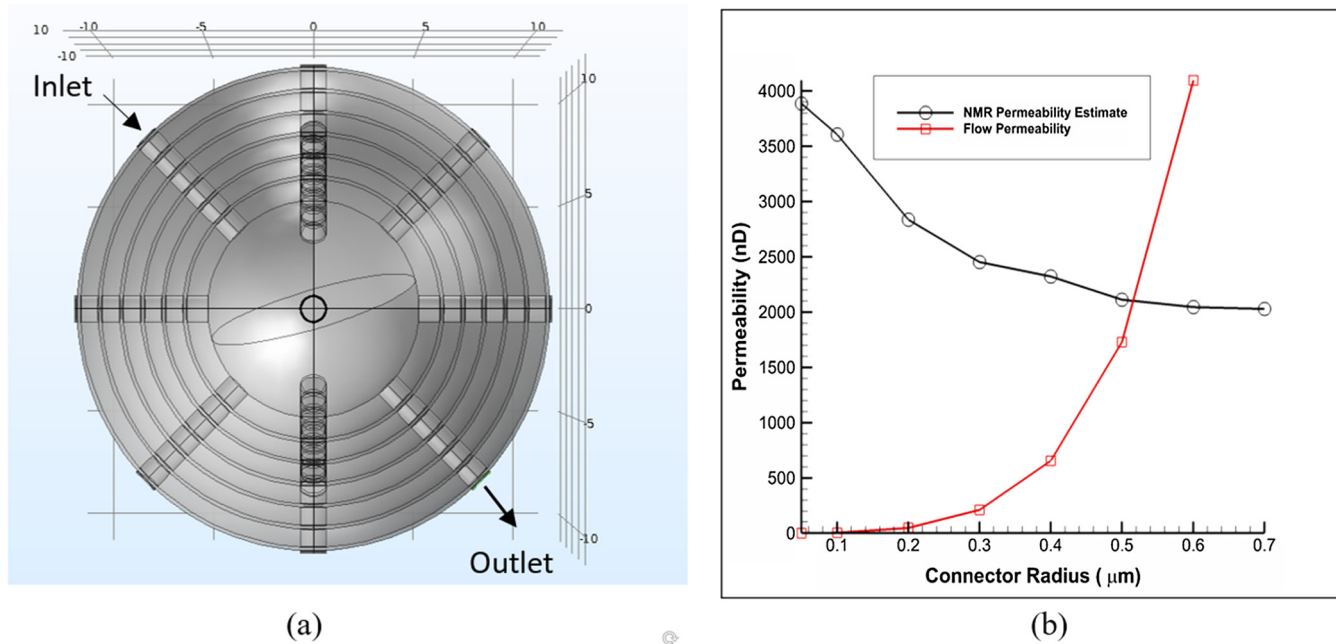


Fig. 9. In (a) the geometry is shown where we solve for the Navier-Stokes equation with the pressure inlet and outlet shown. A tilted disc is included at the center. (b) shows the permeability calculated from the flow with different connectors, and this is compared with the permeability estimates with NMR.

From the previous theoretical discussion of the two pore system and Fig. 4, we know that at the beginning, the small pores start to relax fast reducing in the number of excited protons (concentration) while the big pore is relaxing slow. Although this discussion was in a 1-D context, in the 3-D scenario there are a lot more pathways that affect the trend in the T_2 distribution curve.

In the current model, this trend occurs first as if no connection exists between the pores. Once there is a substantial difference in the concentration of the small and big pores, then upon the presence of connectors, there is a supply of flow from the big pore, which does not allow any further drop in amplitude within the small pores until a balance is reached. Part of the particles in the small pores relaxes fast under the conditions that the supply from the big pore is unimportant. After this, the remaining excited protons will begin to relax together with those in the big pore. Therefore, not much shift in T_2 of the small pores is observed. However, as there is supply from the big pore, there is more pores accessed and the relaxation time of the big pore begins to decrease substantially. It is shown that for good enough connectors we arrive at a single peak amplitude at $T_2 = 25$ ms. This rapid decrease in T_2 in the big pore causes the permeability estimate to give a decrease in permeability for better connectors.

In real rocks, this type of scenario can occur in multiscale regions for example when you have a large pore surrounded by many smaller sized pores. This could be the case in carbonates or in shales. In shales pores can be surrounded by organic matter with ultra-small pores that can lead to this behavior in the T_2 distribution. Also in shales, a lot of times there are either no connections between pores or there are weak connections. In these cases the NMR distribution curves show small pores relaxing fast and big pores relaxing slow with multiple peaks. However, when there is supply from the big pore to the small pores, then we would observe a similar trend as just described where there is a big shift in T_2 of the large pores. Therefore, the above NMR estimate for permeability can result in a decreasing trend as in Fig. 9. Different types of rock patterns which do not contain such multi-scales of grain sizes will satisfy a consistent trend with the NMR permeability estimate.

In the section below a virtual porous medium is constructed from the given grain size distribution as input; and the NMR T_2 distribution curve is compared to the experimental results. First we demonstrate that indeed the NMR T_2 geometric mean T_{2gm} , has a decreasing trend as we reduce the permeability by dilating the grains. This is consistent with the NMR permeability estimate given above. The entire point of the dilation of the grains though is to try to mimic a real rock formation. As will be seen, the numerical simulations of NMR performed on virtual porous media patterns do not lead to a wide distribution typical of experiments of NMR. However, through dilating and rescaling the size of the grains a wider distribution matching the experimental results are obtained.

6. NMR predictions for natural porous media patterns

The above numerical models were used to understand the effect of connectors on the NMR measurements. In this section we incorporate a computational physics approach in order to obtain the NMR relaxation and T_2 distribution for natural porous media patterns. The porous media used in the sample is a Lane Mountain Silica sand pack with a sieve size of 50 (LM50), where experimental results giving the NMR T_2 distribution is performed and compared and used as a comparison to the numerical simulation.

In the numerical approaches the accuracy of the numerical method depends on factors such as the mesh and the type of discretization scheme used. However, scanned cross-sectional images from rocks are pixelated and the geometry of the pore structure is limited in detail and accuracy through the resolution. In computational rock physics or sometimes referred to as digital core analysis (DCA) it has become a trend to treat each voxel as a mesh element because these structured meshes are simpler to implement and they lead to high quality meshes. Unfortunately, these elements do not conform to the geometry and result in an increase in the surface area. While many physical properties of flow within pore structures are predicted using voxel based meshing, for the numerical prediction of NMR signatures the extra surface area will mimic

the effect of roughness and will affect the T_2 distribution. More discussion on the effect of voxelized representation on NMR simulations is given elsewhere [5].

Constructing a natural porous medium pattern for numerical studies requires actual scanned images of rock. This could be CT or micro-CT scans, or high resolution FIB-SEM scans depending on the type of modeling we are interested in. Obtaining the images with the required resolution is ideal in order to determine the physical properties of the porous media and for DCA. However, if scanned images of a sample are unavailable, one can use the large inventory of particle (grain) size distribution (PSD) data to mathematically construct the porous media that will represent the sand pack. In this work, such a physics based modeling approach is used to generate a virtual porous media pattern based on the PSD data [17]. This constructed virtual porous media that is based on a cubic packing of spheres, is generated from the input given by the PSD data of the LM50. This is depicted in Fig. 10, where sides of the box are 1.63 mm. Our mathematical model is the Bloch-Torrey equation where it is imposed that the relaxivity at the walls of the pores to be $\sigma_s = 25 \mu\text{m/s}$ for sand, and the bulk relaxation time to be 2.0 s. Furthermore we take the diffusion constant to be $D = 2.07 \times 10^{-9} \text{ m}^2/\text{s}$ as we did in the previous models. The governing equation is solved numerically using COMSOL Multiphysics which incorporates the Finite Element discretization scheme. From the previous discussion, we use an unstructured tetrahedral mesh that conforms to the geometry as seen in Fig. 10(b).

The spheres are representations of actual grains, although, actual grains are not spherical and their points of contact are not points but are prolonged areas that overlap. In order to account for this we dilate the spheres in Fig. 10 in order to represent packings that are somewhat non-spherical. This procedure makes the T_2 distribution wider because it cuts the connectivity. We point out that in real sands rather than the artificial ones used in the laboratory, there is also the process of cementation that would widen the NMR spectrum even further and thus the practitioner should perform the simulations using more realistic models that account for this effect. Fig. 11 demonstrates the dilation procedure for a single cross-section of the X-Y plane. In Table 1 we make a

Table 1

Gives tabulated values for the experimental, and the numerical simulation of the NMR distribution curve of the original packing, and the dilated packing. The normalized amplitude is given as the amplitude divided by the total amplitude multiplied by the porosity, so that the sum of the amplitudes give porosity. The effect of dilation decreases T_{2gm} which results in a decreasing trend for the NMR estimate for permeability. In this case, the estimate is consistent with actual permeability calculations.

	Porosity (%)	Absolute permeability (D)	T_{2gm} (ms)
Experimental	35.75	2.4	444.7
Virtual PM (without dilation)	46.40	19	666.4
Virtual PM (with dilation)	30.0	4.6	521

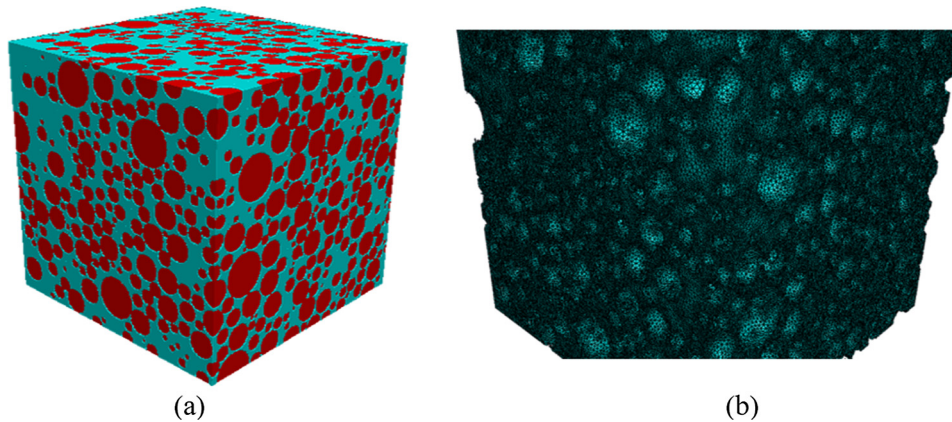


Fig. 10. (a) Depicts the 3D reconstructed packing from grain size distribution data. In (b) we demonstrate the unstructured tetrahedral mesh used. In (b) we have omitted the grains in the computational domain because we are only interested in simulating the diffusion process in the pores and by doing this we save on computational expense. The porosity here of the packing is 0.464.

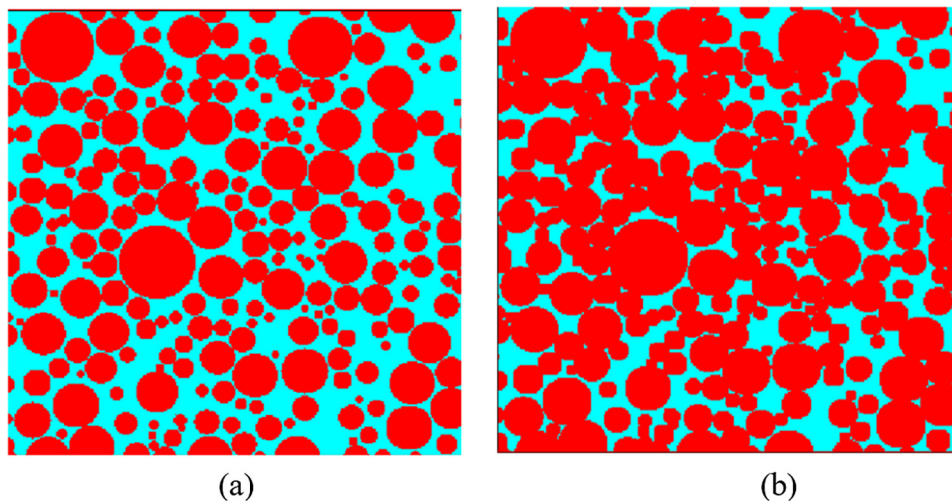


Fig. 11. In (a) we show a single cross section in the X-Y plane of the original packing, and (b) depicts the case where we incorporate dilation of the grains on the same slice.

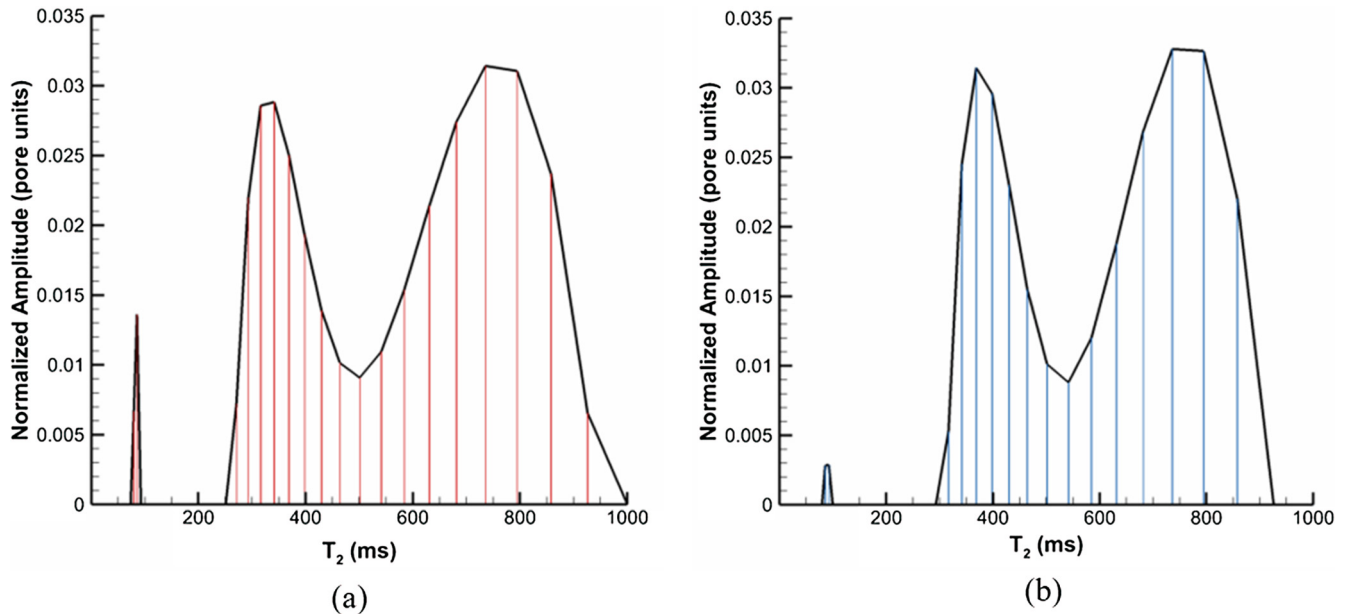


Fig. 12. The laboratory experimental result for the T_2 distribution is shown in (a). In (b) we show the numerical simulation results for the packing of spheres where we have incorporated slight dilation of the grains. Here relatively good agreement is shown where the relaxation times for the different peaks show close proximity.

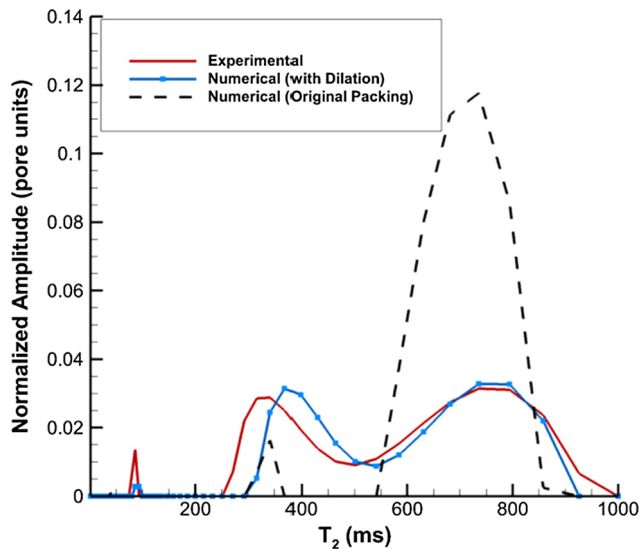


Fig. 13. This figure compares the experimental with the numerical results with and without the dilation. Without dilation, more flow occurs and most of the concentration is represented by one big peak. However, dilation attempts to mimic the more realistic non-spherical features of the artificial sand packs. Its effect is that it cuts the connection and produces extra peaks and comparing well with the experimental results.

comparison of the geometric mean of T_2 (T_{2gm}) between the case without dilation and the dilated grains. This is then compared with the experiment. It is shown that the effect of dilation leads to a decreasing trend for NMR estimate of permeability that is consistent with actual flow permeability calculations. We compare the NMR T_2 distribution of the experimental result in Fig. 12(a) with the numerical result for the case where we dilated the spherical grains as shown in Fig. 12(b). Fig. 13 shows the comparison between the experimental, the numerical with dilation, and the original packing. It is seen here that as the original packing has more inter-connected pore space, the numerical results will be mainly concentrated in a single peak. However the dilation of the

grains is incorporated in order to mimic the true sand packs and as is observed, a wider spectrum is obtained which better matches the experimental results. The amplitudes are normalized by dividing by the total amplitude and multiplying by the porosity.

7. NMR predictions for scanned images

In this section we do a comparison of the NMR T_2 distribution obtained for a sandstone core plug, with the NMR T_2 distribution obtained from numerical simulations. The computational method uses as input a digitized 3D reconstructed image of a piece of the sandstone that is obtained from a set of μ -CT scanned cross-sectional images. Fig. 14 depicts different cross-sections of the 3D sample. We have obtained 635 cross-sectional images of the sandstone where its dimensions are given by $512 \times 512 \times 635$ voxels ($3.18 \times 3.18 \times 3.94$ mm) as shown in Fig. 15.

There are several issues that need to be considered when numerically simulating NMR diffusion process from real images of rocks. NMR T_2 response depends strongly on the pore structure which includes the number of pores (and grains and other minerals) of the sample, the size of pores, and the surface roughness of the grains. Typically, the sample should be large enough to cover a wider range of pore sizes so that under the fast diffusion regime some of the excited protons will relax independently within pores. Otherwise, as discussed in the previous sections we would obtain a single peak due to the interconnectivity between the pores.

Identifying and highlighting (or masking) the different grey scale of the image (which correspond to pores and other grains and minerals) is the thresholding process. There is a wide range of grey scale in the image, however after identifying the pore space, the thresholding is adjusted to cover those spaces stopping prior to generating any noise content in the image. An example of thresholding the pore spaces as viewed in a 2D cross-section of the image is shown in Fig. 14(d). Thus, the size and number of the pores is greatly influenced by the level of sophistication in the thresholding process. This is in contrast to the virtual porous media patterns generated in the last section where the thresholding is very straightforward as the pores and grains are constructed and identified through the physics based modeling.

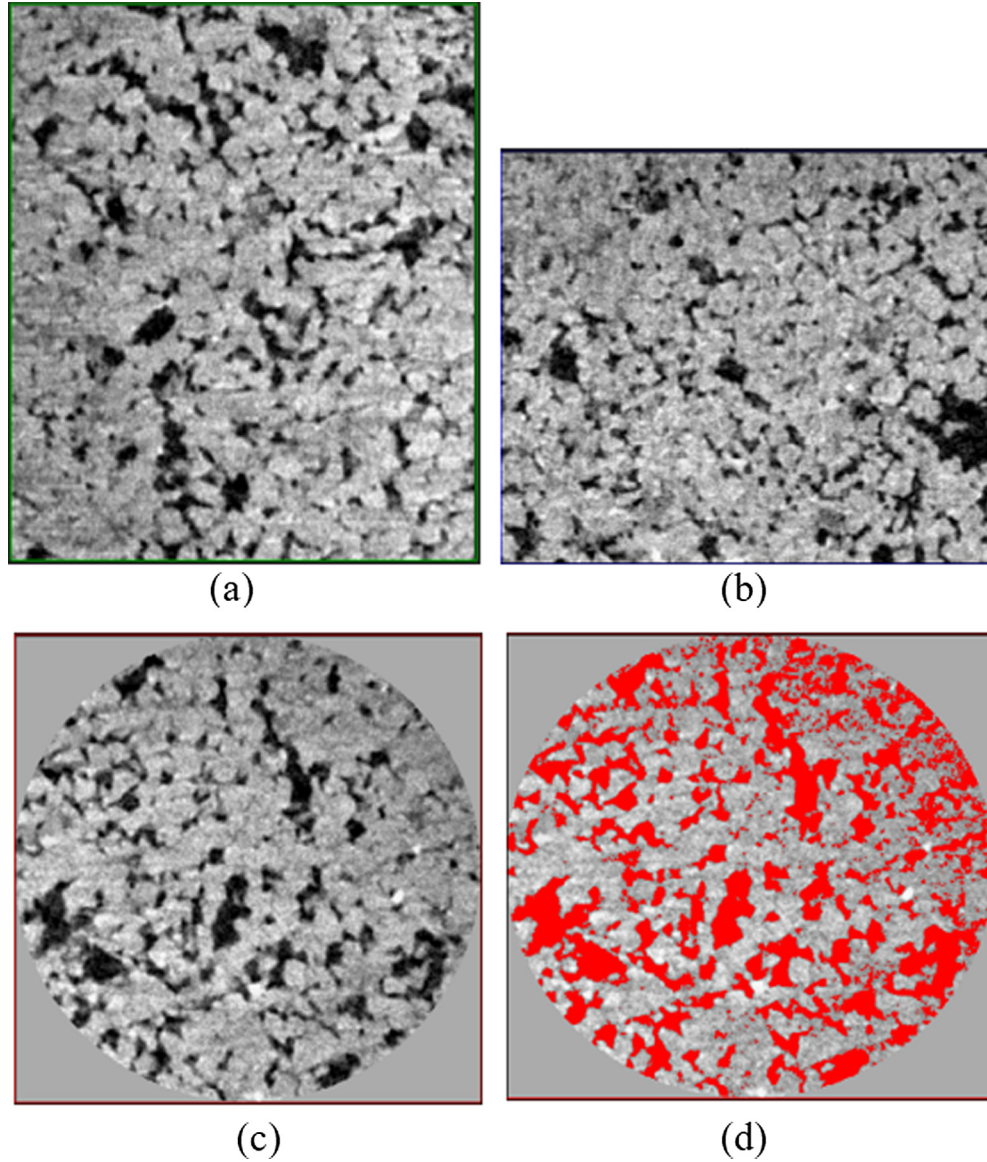


Fig. 14. μ -CT scans of sandstone. (a) is the YZ plane, (b) is the XZ plane, and (c) is in the XY plane. (d) Represents the cross-section of (c) where the pores are masked using a thresholding technique.

In order to account for surface roughness, in the real rock images, high resolution (typically at the Nano-scale) scanned images are required. Even if these high resolution images are obtained, from for example FIB-SEM scans, then the problem of meshing the geometry and the computational expense associated with it can be a formidable task. The μ -CT scans that are obtained here have a lower resolution ($\sim 10 \mu\text{m}$) which effectively smooths the surface and hides the roughness. Therefore, in order to represent the rough surface with a smooth surface we must consider an effective surface relaxivity. Several authors have used the fractal approach to model and characterize surface roughness (see for example [19,13]). We will not cover this modeling approach here as it is beyond the subject of this paper. However, we emphasize that it is crucial to incorporate surface roughness into the model as it is the roughness that influences the effective surface relaxivity thus affecting the NMR T_2 distribution. The typical surface relaxivity of sandstone is about $20 \mu\text{m/s}$ as referenced in [9], and in [15]. Here to account for the rough surface, for this case, we take an effective surface relaxivity of $100 \mu\text{m/s}$. We point out also that with

a resolution of $10 \mu\text{m}$ and a surface relaxivity of $20 \mu\text{m/s}$ for sandstone, we have:

$$\frac{1}{T_2} = \sigma_s \frac{S}{V} = 20 \frac{\mu\text{m}}{\text{s}} \times \frac{3}{10 \mu\text{m}} = 6 \frac{1}{\text{s}}.$$

Thus the T_2 corresponding to this resolution will be approximately 167 ms. In Fig. 16 we compare the NMR T_2 distribution for both the experiment and numerical results and we obtain a very good match except at lower values of T_2 (lower than 167 ms), where the results will not match due to the resolution of the μ -CT scans. We further normalized the amplitudes with respect to the total amplitude of the two largest peaks where the numerical results are deemed to be reliable.

From the computational point of view we make the final remark that in meshing the geometry, voxel based meshes are fast and give good quality meshes, but they generate a jagged step like surface which can give the effect of roughness, and can be misleading as it is a consequence of the numerics. In our meshing procedure

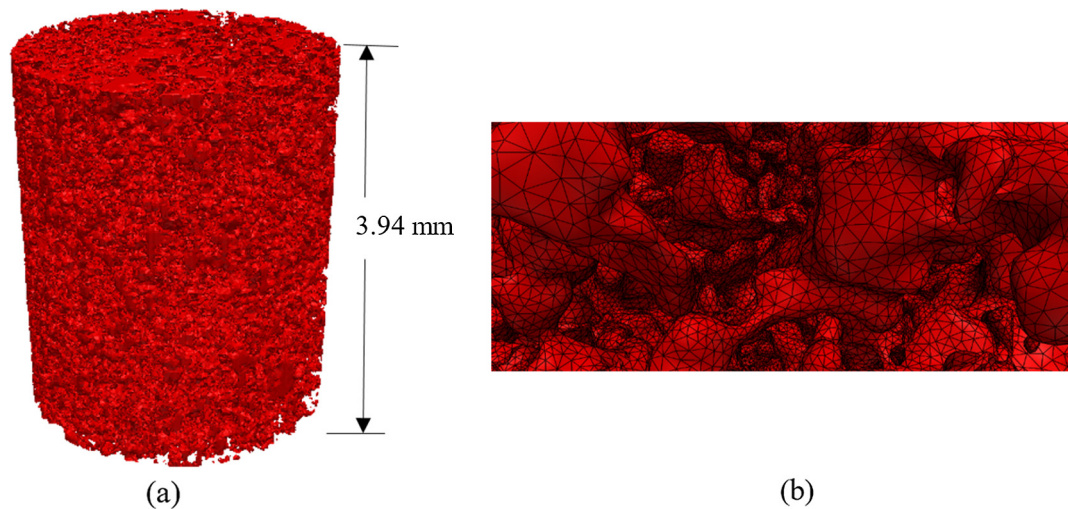


Fig. 15. (a) reconstructed piece from sandstone consisting of 635 cross-sectional images of μ -CT scans. (b) Represents the computational mesh that is invoked, consisting of nearly 49 million tetrahedral cells. Tetrahedrons conform to the shape of the geometry and provides a smooth surface, not the jagged step like surfaces that are due to a voxel based mesh.

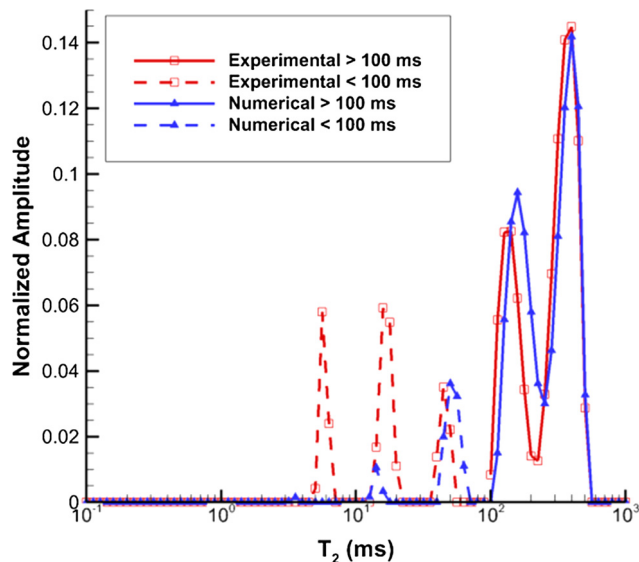


Fig. 16. T_2 distribution comparing both the experimental results (in red) with the numerical simulation (in blue). The S/N ratio in the experiment is 280. The dashed lines represent areas where the numerical simulation may fail to predict the correct T_2 due to low resolution of the μ -CT. Here the amplitudes are normalized with respect to the total amplitude of the two largest peaks (>100 ms). The surface relaxivity in the numerical simulation is taken to be $100 \mu\text{m/s}$ and the bulk relaxation is taken to be 2.5 s with diffusion coefficient of $2.07 \times 10^{-9} \text{ m}^2/\text{s}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

we incorporate a tetrahedral mesh algorithm that conforms to the geometry using Simpleware's ScanIp. This is shown in Fig. 15(b).

8. Conclusions

In this paper we attempt to explain the effect of pore connectivity on the NMR T_2 distribution through theoretical examples, numerical models, and more realistic porous media patterns. The results are consistent and demonstrate that although NMR is known to give information on the pore body size distribution, the effect of pore throats influences the NMR measurement data dramatically and must be taken into consideration. Moreover,

understanding how the pores and connectors contribute to the T_2 distribution will allow for better estimates of permeability.

For tight formations, one can argue that the connectors are small and thus the NMR T_2 measurement is close to the case where there are no pore connectivity. However, in many numerical simulations one must understand that the inter-connectivity through pores, will lead to a mono-exponential decay that will in turn produce single peak. If the practitioner is using a virtual porous media pattern as a computational models for study, it is apparent that the geometric structure created through modeling should incorporate all the physical features such as cementation and grain shape into account. On the other hand, if the practitioner has available scanned images of the actual rock, then the computational model used will be the reconstructed binarized image. In this case other factors such as voxel based meshing can lead to misleading results as they mimic roughness and thus in turn affect the T_2 and proper interpretations. Although resolution is always a limiting factor, for NMR smooth based meshes conforming to the shape of the geometry should be used, and proper modeling of roughness should be incorporated.

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